



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Pressure

1.8

STUDENT NAME



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PHYSICS WITH
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SYLLABUS OUTLINE:

1.8 Pressure

- 1 Define pressure as force per unit area; recall and use the equation

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$p = \frac{F}{A}$$

- 2 Describe how pressure varies with force and area in the context of everyday examples
- 3 State that the pressure at a surface produces a force in a direction at right angles to the surface and describe an experiment to show this
- 4 Describe how the height of a liquid column in a liquid barometer may be used to determine the atmospheric pressure
- 5 Describe, quantitatively, how the pressure beneath the surface of a liquid changes with depth and density of the liquid
- 6 Recall and use the equation for the change in pressure beneath the surface of a liquid
- change in pressure = density $\times$ gravitational field strength $\times$ change in height

$$\Delta p = \rho g \Delta h$$

PRESSURE:

"Pressure is the force per unit area."

The S.I unit of Pressure is Pascal (Pa) or $\frac{\text{N}}{\text{m}^2}$.

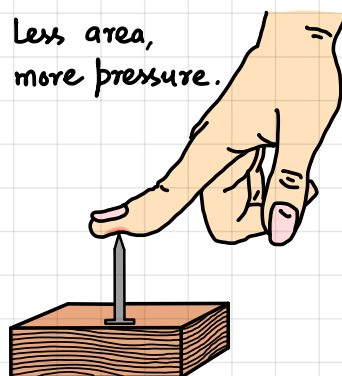
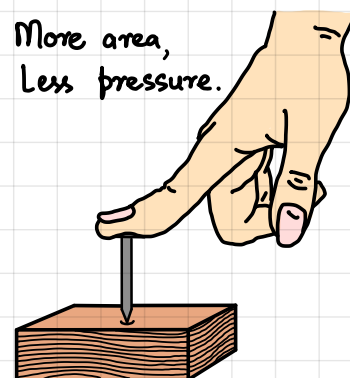
FORMULA:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$P = \frac{F}{A}$$

FACTORS:

- Area: $P \propto \frac{1}{A}$
- Force: $P \propto F$



ATMOSPHERIC PRESSURE.

Air particles are continuously moving in random direction in a straight line hence collide with each other, exert force on a given area. This is why atmosphere has pressure. The value of atmospheric pressure at the sea level is $100,000 \text{ Pa}$. Atmospheric pressure decreases with height.

BAROMETER:

"A device which is used to measure the pressure of atmosphere is called barometer."

Pressure at A = Pressure at B
Value of atmospheric pressure at sea level is $1.0 \times 10^5 \text{ Pa}$ which makes the mercury rise by 76 cm.

Q Calculate atmospheric pressure at the summit of mountain where mercury rises by 40 cm.

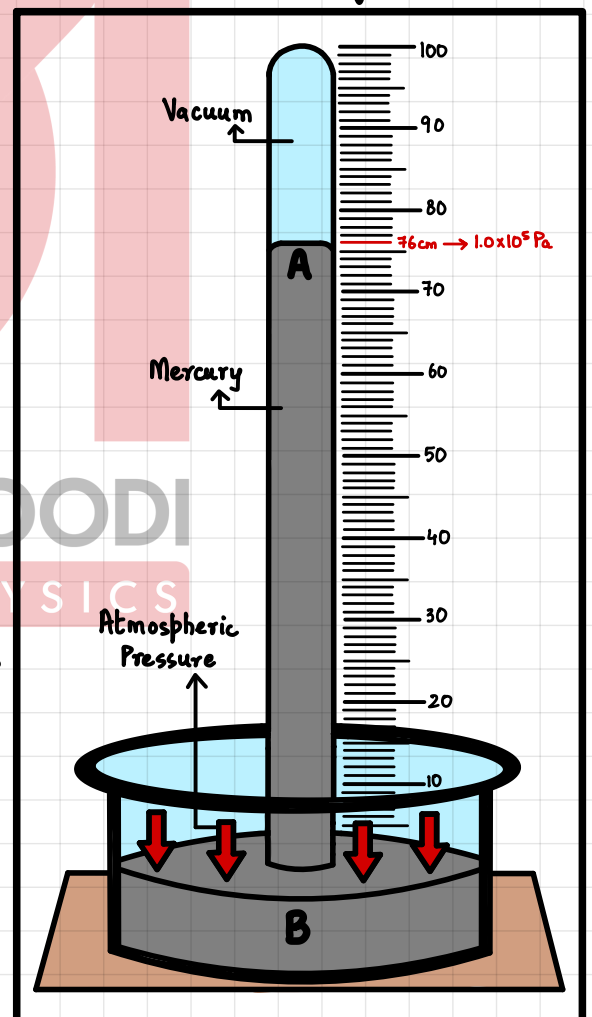
At sea level :

$$76 \text{ cm} \rightarrow 1.0 \times 10^5 \text{ Pa}$$

$$40 \text{ cm} \rightarrow x$$

$$x = \frac{1.0 \times 10^5 \times 40}{76}$$

$$x = 0.526 \times 10^5 \text{ Pa}$$

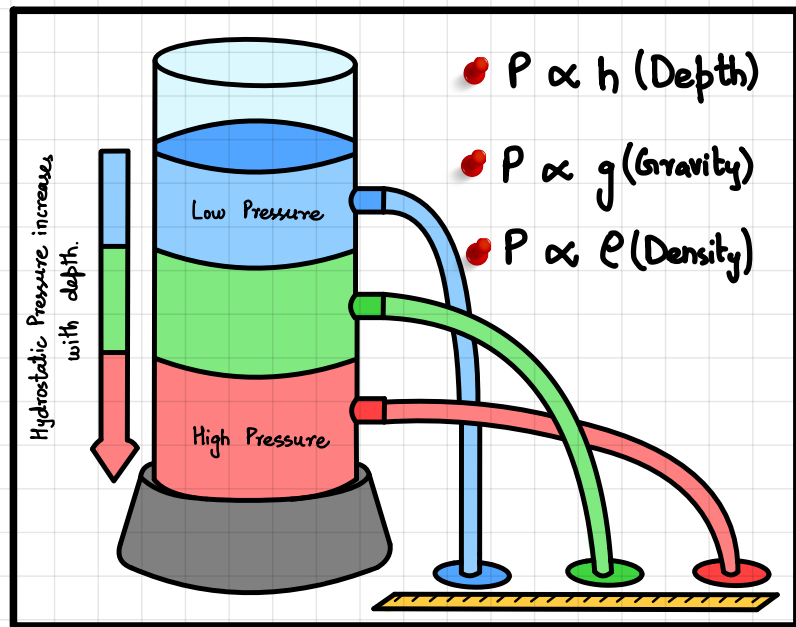


HYDROSTATIC PRESSURE:

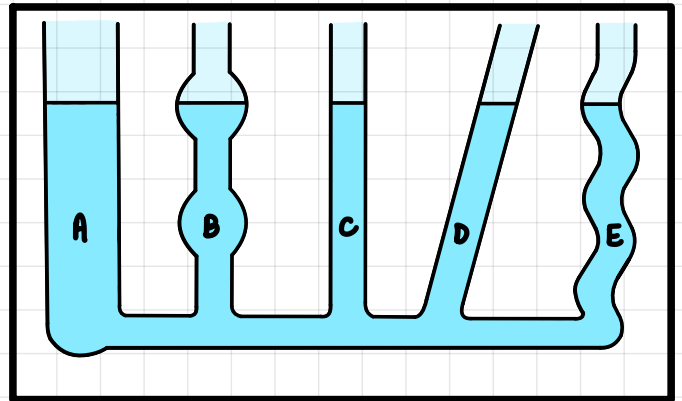
"Pressure exerted by liquids is called hydrostatic pressure."

FORMULA:

$$P = \rho g h$$



In this diagram, hydrostatic pressure remains same in all the flasks as it does not depend on shape of flask.





MATCH THE FOLLOWING



Pressure is defined as

Pressure is inversely proportional

Hydrostatic pressure is given by:

Hydrostatic pressure increases

When pressure is applied,

The reason of atmospheric pressure

Atmospheric pressure decreases

Value of atmospheric pressure

to surface area of object.

force per unit area.

when depth increases.

$$P = \rho g h.$$

is the air particles moving randomly, hence colliding with each other.

object always goes from high to low pressure region.

at sea level is $1.0 \times 10^5 \text{ Pa}$.

if altitude is increased.